

Programme	B. Sc. Computer Science				
Course Title	Machine Learning Algorithms				
Type of Course	Elective				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Understanding of basic mathematics and statistics (linear algebra, calculus, probability)				
Course Summary	This course introduces the fundamental concepts, algorithms, and applications of machine learning				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand basic concepts of machine learning, including supervised learning, unsupervised learning, and reinforcement learning	U	C	Instructor-created exams / Quiz
CO2	Understand the mathematical foundations of machine learning algorithms, including concepts such as optimization, linear algebra, probability, and statistics	U	C	Assignment / Seminar presentations/ Exams

CO3	Demonstrate proficiency in various machine learning algorithms, such as linear regression, logistic regression, decision trees, support vector machines, k-nearest neighbors, clustering algorithms, and neural network	U	P	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO4	Explore techniques for feature engineering and feature selection to improve the performance of machine learning models.	U	P	Instructor- created exams / Home Assignments
CO5	Evaluate machine learning models using appropriate metrics and techniques, including cross-validation, precision, recall, F1 score, ROC curves, and confusion matrices.	Ap	P	Writing assignments/ Exams/ Seminar Presentations
CO6	Develop critical thinking skills to analyze and solve complex problems using machine learning approaches.	Ap	P	Case Study/ Group discussions/ Presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Mathematical Foundation for Machine learning		14	20
	1	Introduction to key concepts: features, labels, training, and testing	2	
	2	Designing a Learning system	1	
	3	Types of learning; supervised, unsupervised and reinforcement	2	
	4	Introduction to linear algebra- Vector :-Vector operations: addition, subtraction, scalar multiplication	2	
	5	Matrices- Matrix operations	2	
	6	Eigenvalues and Eigenvectors	2	
	7	Foundations of Probability for ML:- Introduction to probability	1	
II	8	Random Variable, Probability distributions (Normal and gaussian-basics only), Naïve bayes	2	
	Feature Engineering and Preprocessing		12	15
	9	Data Preprocessing and Feature Engineering: Data Representation, Data Preprocessing	2	
	10	Features and Types	3	
	11	Dimensionality Reduction – Feature Identification	2	
	12	Feature selection	2	
III	13	Feature extraction - Feature Importance	3	
	Regression and Classification		12	20
	14	Regression: Linear Regression – Non-Linear regression	2	
	15	Evaluation metrics for regression	1	

	16	Classification: Binary, multi-class, and multi-label classification	1	
	17	lazy learners- (KNN) - tree-based techniques (Decision Tree)- kernel based techniques (SVM) - probabilistic techniques (Naïve bayes)- and ensembled techniques (bagging, boosting, voting)	7	
	18	Evaluation metrics for classification.	1	
IV	Clustering and Rule Mining		10	15
	19	Clustering: Partitioning based (K Means)	2	
	20	Hierarchical based (Divisive)	2	
	21	Rule mining: Apriori algorithm, FB Growth - association rules.	4	
	22	Outlier Detection - LOF	2	
V	Open Ended Module		12	
	1	Ethical considerations in machine learning	3	
	2	McCulloch-Pitts neurons, Hebb's networks	3	
	3	Hopfield networks, Boltzmann machines	2	
	4	Reinforcement Learning: Markov Decision Processes (MDPs), Q-learning.	4	

References

- Ethem Alpaydin, Introduction to Machine Learning- 3rd Edition, PHI
- Machine Learning by Mitchell, Tom M. (Tom Michael), McGraw-Hill
- Mathematics For Machine Learning, Marc Peter Deisenroth A. Aldo Faisal Cheng Soon Ong
- "Pattern Recognition and Machine Learning" by Christopher M. Bishop.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	1	-	-	1	-						
CO 2	3	-	-	-	1	-						
CO 3	1	3	1	1	2	3						
CO 4	1	-	1	1	2	3						
CO 5	1	-	-	-	2	3						
CO 6	1	2	2	2	3	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium

3	Substantial / High
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Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4	✓	✓	✓
CO 5	✓	✓	✓
CO 6		✓	